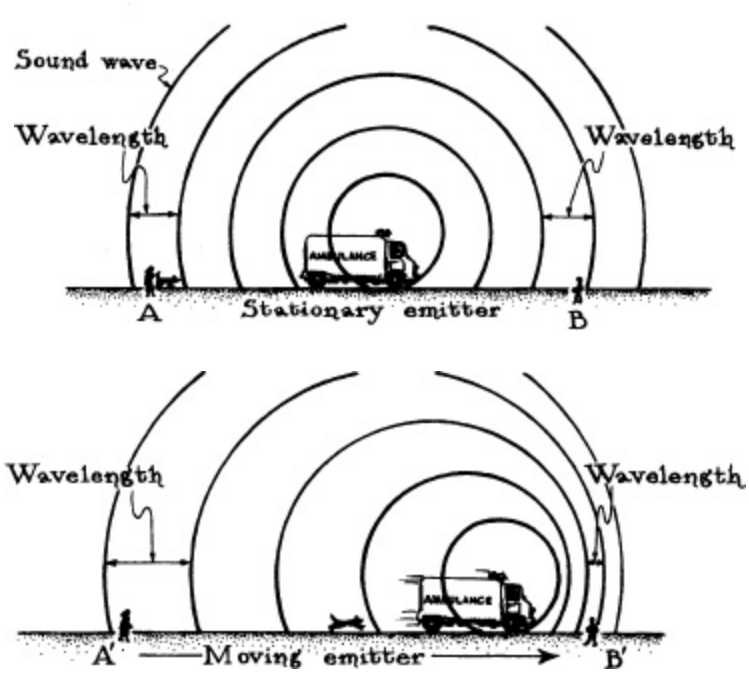
Box 2.3

Doppler Shift

Whenever an emitter and a receiver of waves are moving toward each other, the receiver sees the waves shifted to higher frequency—that is, shorter period and shorter wavelength. If the emitter and receiver are moving apart, then the receiver sees the waves shifted to lower frequency—that is, longer period and longer wavelength. This is called the *Doppler shift,* and it is a property of all types of waves: sound waves, waves on water, electromagnetic waves, and so forth.

For sound waves, the Doppler shift is a familiar everyday phenomenon. One hears it in the sudden lowering of the sound's pitch when a speeding ambulance passes with siren screeching (drawing b), or when a landing airplane passes overhead. One can understand the Doppler shift by thinking about the diagrams below.



What is true of waves is also true of pulses. If the emitter transmits regularly spaced pulses of light (or of anything else), then the receiver, as the emitter moves toward it, will encounter the pulses at a higher frequency